

ÁLVARO ZARZUELA

SECURITY & SOFTWARE ENGINEER

Final-year Software Engineering student focused on cybersecurity and network defense. As founder and lead developer of Valiware Loyalty, I apply cryptography and modern architectures to solve real-world problems. Analytical and self-driven, I prioritize reliability and the delivery of secure, high-impact solutions.

CONTACT

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PORTFOLIO / GITHUB:
View Portfolio /
Github - Álvaro Zarzuela

LOCATION:
Huelva, Spain (Available for remote)

ADDITIONAL TRAINING

IBM SkillsBuild

- Cybersecurity Certificate, in progress.

Hack4u Academy

- Hands-on training in vulnerable environment setup and guided exploitation, in progress.

CADE (Andalucía Emprende)

- Business incubation, mentorship, and commercial validation program | 2026

LANGUAGES

- **Spanish - Native**
- **English - B2**
  360° LPT Certificate, 2026

EXPERIENCE AND TECHNICAL PROJECTS

Founder & Lead Developer - Valiware Loyalty
2026 – Present | CADE (Andalucía Emprende)

- Developed an end-to-end SaaS platform and high-concurrency serverless backend (Next.js, TypeScript, PostgreSQL, Redis).
- Integrated NTAG 424 DNA cryptographic NFC chips (rolling dynamic tokens) to enforce anti-fraud physical verification.
- Designed a zero-friction onboarding flow using phone authentication and local persistence.

Cybersecurity Participant / Competitor
Cyber BotCamp Málaga | 2026

- Solved CTF challenges using ethical hacking techniques, system hardening, and network traffic analysis.
- Attended hands-on technical workshops on network auditing, vulnerability assessment, and incident response.

Cybersecurity Labs & Hands-On Auditing
Independent Projects / Hacking Platforms

- Solved vulnerable virtual machines on TryHackMe and Hack The Box; configured and hardened specialized Linux environments (Parrot OS).
- Automated reconnaissance, port scanning, and vulnerability assessment routines using Python and Bash scripts.

SKILLS & TECHNOLOGIES

- **Development & Architecture:** TypeScript, Next.js, Node.js, REST APIs, Git.
- **Databases & Cloud:** PostgreSQL (Supabase), Redis, Serverless Architectures.
- **Cybersecurity & Cryptography:** Applied Cryptography (NFC NTAG 424 DNA), Ethical Hacking, Vulnerability Assessment, Wireshark, Burp Suite, Nmap.
- **Systems & Networks:** Linux (Parrot OS, Ubuntu), Windows Server, TCP/IP Protocols, Traffic & Log Analysis.
- **Programming Languages:** TypeScript, Python, Java, C++, Bash, SQL.
- **Core Competencies:** Analytical problem solving under constraints, rapid self-directed learning, and technical teamwork.

EDUCATION

Computer Engineering

- *University of Huelva* (2023 - 2027)

Science Baccalaureate

- *IES Hozgarganta* (2021 - 2023)

- Participated in hands-on workshops on vulnerability analysis, defensive security, and network auditing.
- Solved Capture The Flag (CTF) challenges, applying ethical hacking and system hardening techniques.
- Analyzed network traffic and practiced incident response in controlled test environments.